

PHASE 1: INGESTION & PII REDACTION

Data Ingestion & Regex-NER Engine

Detects SSN, Email, Phone, IP, CC, API Keys

Token Masking & In-Memory Queue

Volatile Buffer (/dev/shm) | 3-Pass DoD Wipe

✓ Clean Tokens

PHASE 2: LOCAL ALTERNATING DP-LORA

Base LLM W_0 (Frozen) + LoRA (A, B)

Trainable Rank $r=8$ (0.144% active params)

LA-LoRA Alternating DP-SGD Update

Step 2k-1: Update A | Step 2k: Update B

Differential Privacy & RDP Accountant

Gaussian Noise $N(0, \sigma^2 C^2 I)$ | (ϵ, δ) -DP ($\epsilon < 8.0$)

✓ LoRA Tensors

PHASE 3: HW BINDING & AES-256-GCM

Hardware Probe & Fingerprint (F)

$F = \text{SHA-256}(\text{CPU_Microcode} \parallel \text{UUID} \parallel \text{Disk})$

HKDF Key Derivation & AES-256-GCM

$K_{\text{master}} = \text{HKDF}(F, s)$ | Up to 4,602 MB/s Enc

RSA-2048-PSS Signed Package

Header Sign $\sigma_H \rightarrow$ Artifact: adapter.seclora

✓ adapter.seclora

PHASE 4: ENCLAVE RUNTIME DEPLOYMENT

8-Gate Sequential Verification Engine

Header, RSA Sig, HWID Match, GCM Tag Check

Docker CGroup Execution Enclave

Volatile VRAM Allocation | 1.33 ms Latency